

STATEMENT OF WORK
Title: Service Contract for Optical Frequency Comb System

I. BACKGROUND INFORMATION

In FY22, NIST purchased an optical frequency comb system. The warranty on that system has now expired. We would like to purchase a service agreement to continue to support this system.

II. PURPOSE

The Neutral Atom Optical Clocks Division at NIST requires the purchase of a one-year service agreement for the optical frequency comb with two option years. The service agreement will include prioritized remote and onsite support, including travel and labor, to resolve technical issues impeding normal operation of the comb system.

PLACE OF PERFORMANCE

Service will begin with remote support, followed by onsite support (at Boulder NIST) as required.

Regular Business Hours. Regular business hours are Monday through Friday, 8:00 am to 5:00 pm Mountain Time, excluding Federal holidays and NIST closures.

PERIOD OF PERFORMANCE

Base Period – May 1, 2025, through April 30, 2026
Option Period I - May 1, 2026, through April 30, 2027
Option Period II – May 1, 2027, through April 30, 2028

III. SPECIFIC REQUIREMENTS

Task 1 – Remote Support

The contractor shall:

1. Maintain priority communication with customer, to provide continuous updating of software and firmware.
2. Provide next business day responses to any customer inquiries or notifications of issues with the optical frequency comb system. This response will be provided by a U.S.-based service engineer via email, telephone, or web conference.
3. If customer support cannot be met with U.S.-based service engineer, support will be given within five business days or less from the customer inquiry, via email, telephone, or web conference.

Task 2 – Repair service

If a customer issue cannot be resolved via remote support, the contractor shall:

1. Provide an onsite visit within five business days after customer notification/inquiry from a U.S.-based service engineer for field diagnosis or repair.
2. For repairs that can be resolved onsite using spare parts already purchased by the customer, repair time is guaranteed in less than four weeks. In this case, the service agreement covers up to 60 hours of repair time.
3. If a repair requires additional hardware or needs to be returned, a repair quote will be issued after system evaluation. The service mission will be completed within 6-8 weeks after mission plan submission and customer approval.

IV. GENERAL REQUIREMENTS

Government Furnished Property

NIST-owned Optical Frequency Comb System

NIST-owned backup hardware, including laser pump modules and control electronics.

All property, data and information provided by the Government in the performance of this task remains the property of the Government and shall be surrendered to the Government upon completion or termination of this requirement. Likewise, all deliverables generated under this requirement remain the property of the Government.

Upon award of this requirement, the Government will provide the Contractor with required access to all equipment and resources needed in the performance of this requirement.

Inspection & Acceptance

Performance Tests:

1. NIST will typically test normal operation of the comb system by using it as part of the NIST transportable Yb optical clock. This includes comb and laser frequency stabilization, as well as frequency stability and phase noise analysis in the optical domain and using the Comb 10 GHz output.

NIST may choose at its discretion to forego this part of acceptance testing.

A visual inspection of the equipment will be performed by the NIST POC to identify surface defects or any form of indication that any equipment was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the equipment due to defects and/or nonconformance. The vendor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the equipment.

The Government will test, inspect, and accept or reject the equipment within 7 business working days from the receipt of the repaired equipment unless otherwise indicated above. The Government reserves the right to conduct quality assurance testing to confirm that a given instrument(s) meets the manufacturer's and/or the Government's performance specifications. It is anticipated that the equipment will meet all manufacturer's specifications and/or the Government's performance specifications identified in the most recent operations and maintenance manual for each piece of equipment and/or in this document.

QUALIFICATION REQUIREMENTS (CONTRACTOR PERSONNEL)

Safety: The Contractor employee must be responsible for knowing and complying with all installation safety prevention regulations. Such regulations include, but are not limited to, general safety, fire prevention, and waste disposal.

Security: NIST is a restricted campus. An identification badge is required for access for entry into buildings and is shown to the armed Security Police when entering the campus.

Identification Badges: Contractor employees must comply with NIST identification and access requirements. Each Contractor employee must wear a visible identification badge provided by the NIST Security Office.

V. SCHEDULE OF DELIVERABLES

Define all deliverables and indicate:

- (1) the quantity desired
- (2) the format in which each deliverable is to be supplied
- (3) the delivery date (based on the number of weeks/days)
- (4) the place of delivery.

The context of each deliverable shall be clear. **Keep in mind that every task does not require a deliverable. Deliverables are always tangible items (i.e., report, minutes, etc.)**

Complete the table providing the information for all the tasks identified above.

Deliverable Number	Description	Format	Due Date	Quantity
1	Remote Support	Verbal, web conference or email communication	1 business day for domestic vendor office or 5 business days for foreign vendor office	As needed during the period of performance of the contract
2	Repair service	Physical repair of equipment	4 weeks for repairs done domestically and 6-8 weeks for repairs done at foreign manufacture location	Up to 60 hours of service during the 12-month period. For foreign repair, additional costs may be required, to be quoted in advance of repair.

Standards of Acceptance: The NIST POC or COR shall review all the above deliverables and respond with an acceptance or request for revision email to the Contractor Point of Contact (POC) within 7 days of receipt of deliverable. This section applies to deliverables which must be reviewed and approved for acceptances (e.g., draft plans, drawings, etc.)

VI. PAYMENT SCHEDULE

Define how the contractor will be compensated. If firm fixed price define percentage of payment or amount for each deliverable.

Advance payment is not authorized. The Contractor must invoice in the arrears according to the payment schedule. The Contractor shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice, in accordance with the following schedule:

1. 100% after installation and acceptance by the POC of fully installed system, AND
2. After the successful completion of the testing requirements set forth in this document under section set forth in this document, AND
3. After successful demonstration by the instrumentation that it performs IAW the technical requirements set forth in this document AND
4. After receiving an invoice submitted properly, IAW the purchase order terms and conditions.

NOTE: Partial shipments and partial invoices will not be accepted, unless otherwise requested and accepted by the Contracting Officer prior to award offer. Proposed payment schedules shall be submitted with vendor's response to the RFQ for consideration.

Payment made on a quarterly basis.